

NETWORK ROUTING PROTOCOLS

A Complete Reference Guide

AODV · RIP · OSPF · BGP (eBGP / iBGP)

QUICK PROTOCOL OVERVIEW

Protocol	Type	Algorithm	Scope	Best For
AODV	Reactive / Ad-hoc	Bellman-Ford	MANETs	Mobile / wireless ad-hoc networks
RIP	Distance Vector	Bellman-Ford	Intra-domain	Small LANs, simple IGP
OSPF	Link State	Dijkstra (SPF)	Intra-domain	Enterprise, ISP core IGP
BGP	Path Vector	Best Path Selection	Inter-domain	Internet, multi-AS routing

1. AODV — Ad-hoc On-Demand Distance Vector

What is AODV?

AODV is a **reactive (on-demand) routing protocol** designed for **mobile ad-hoc networks (MANETs)**. Routes are discovered **only when needed** — minimising control overhead in dynamic topologies.

WORKING PHASES

Phase	Description & Example
1. Route Request (RREQ)	Source node S broadcasts a RREQ packet when it needs to reach destination D. <i>Example: Node A → needs to reach Node E. A has no route, so it floods RREQ to all neighbours (B, C).</i>
2. Route Reply (RREP)	Destination D (or an intermediate node with a fresh route to D) sends a unicast RREP back to S. <i>Example: Node E receives RREQ, sends RREP back via D → C → A (reverse path).</i>
3. Route Maintenance (RERR)	If a link breaks mid-session, an RERR (Route Error) packet is sent upstream to invalidate the stale route. <i>Example: If link C-E breaks, C sends RERR to A, which then re-initiates RREQ.</i>
4. Route Expiry	Routes have a lifetime. Unused routes expire; active routes are refreshed automatically.

NETWORK TOPOLOGY EXAMPLE

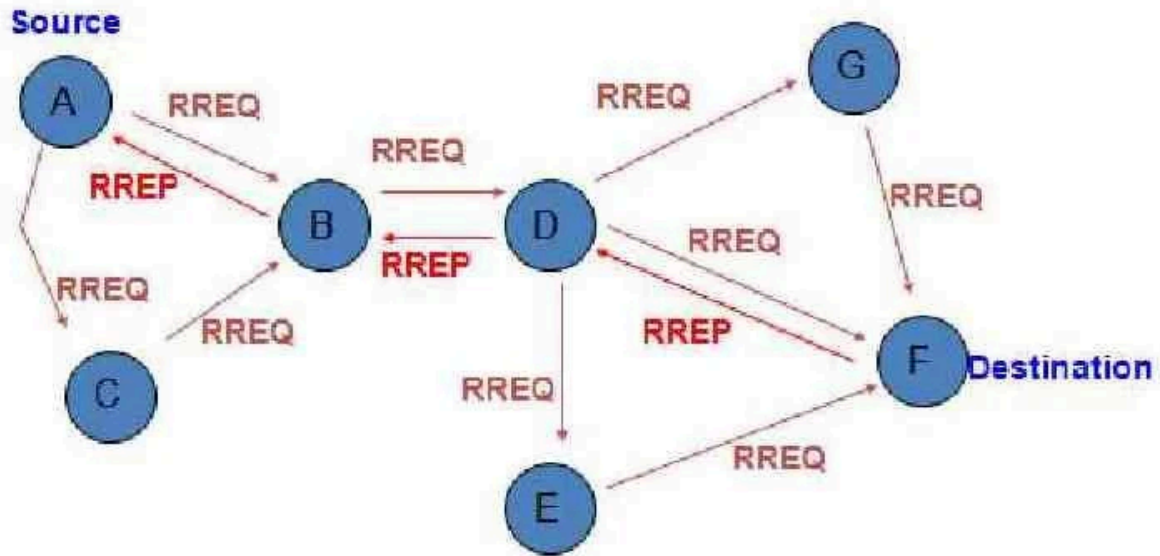
AODV Route Discovery: A → B → D → E

[A] —RREQ—> [B] —RREQ—> [D] —RREQ—> [E]

[A] ←RREP— [B] ←RREP— [D] ←RREP— [E]

★ Route established: A → B → D → E (unicast data flow)

Route Table at A: Dest=E, NextHop=B, HopCount=3, Lifetime=30s



✓ ADVANTAGES

- No periodic route updates → low bandwidth overhead
- Scales well with node mobility
- Loop-free routing guaranteed via sequence numbers
- Supports both unicast and multicast
- Conserves battery in mobile devices

✗ LIMITATIONS

- Initial route discovery adds latency (flood overhead)
- Not suitable for large wired/stable networks
- RREQ flooding can cause broadcast storms
- Route lifetime management is complex
- Security vulnerabilities (e.g., black-hole attacks)

2. RIP — Routing Information Protocol (Distance Vector)

What is RIP?

RIP is an **Interior Gateway Protocol (IGP)** that uses the **Bellman-Ford distance-vector algorithm**. Each router shares its **entire routing table** with directly connected neighbours every **30 seconds**. Metric = hop count. Max hops = **15** (16 = unreachable).

WORKING PHASES

Phase	Description & Example
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1. Initialisation	Each router populates its table with directly connected networks (cost = 1 hop).
2. Update Exchange	Every 30 s, each router sends its full routing table to neighbours (UDP port 520 / 521). <i>Example: Router A tells Router B: 'I can reach Network 10.0.0.0 in 1 hop'.</i>
3. Table Update	Router adds 1 to received hop count. If new path < current metric → update table (Bellman-Ford relaxation). <i>Example: B learns 10.0.0.0 via A = 2 hops. B propagates this to C.</i>
4. Convergence	After multiple 30-second cycles, all routers have consistent tables. Convergence time can be several minutes for large networks.
5. Failure Handling	If a route is not updated for 180 s (timeout), it is marked invalid. Garbage-collection timer (120 s) then removes it. Split Horizon & Poison Reverse prevent count-to-infinity loops.

ROUTING TABLE EXAMPLE (Router B, 4-node topology A–B–C–D)

Destination	Next Hop	Metric (Hops)	Interface
192.168.1.0/24	Direct	1	Gi0/0
192.168.2.0/24	192.168.1.1 (A)	2	Gi0/0
192.168.3.0/24	192.168.4.1 (C)	2	Gi0/1
192.168.4.0/24	192.168.4.1 (C)	3	Gi0/1

✓ ADVANTAGES - Simple to configure and understand - Widely supported on legacy hardware - Low CPU/memory requirements - Good for small, flat networks (<15 hops)	✗ LIMITATIONS - Max 15 hops — cannot scale beyond small networks - Slow convergence → routing loops during failures - Bandwidth-intensive (full table every 30 s) - No support for VLSM in RIPv1 Count-to-infinity problem
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3. OSPF — Open Shortest Path First (Link State)

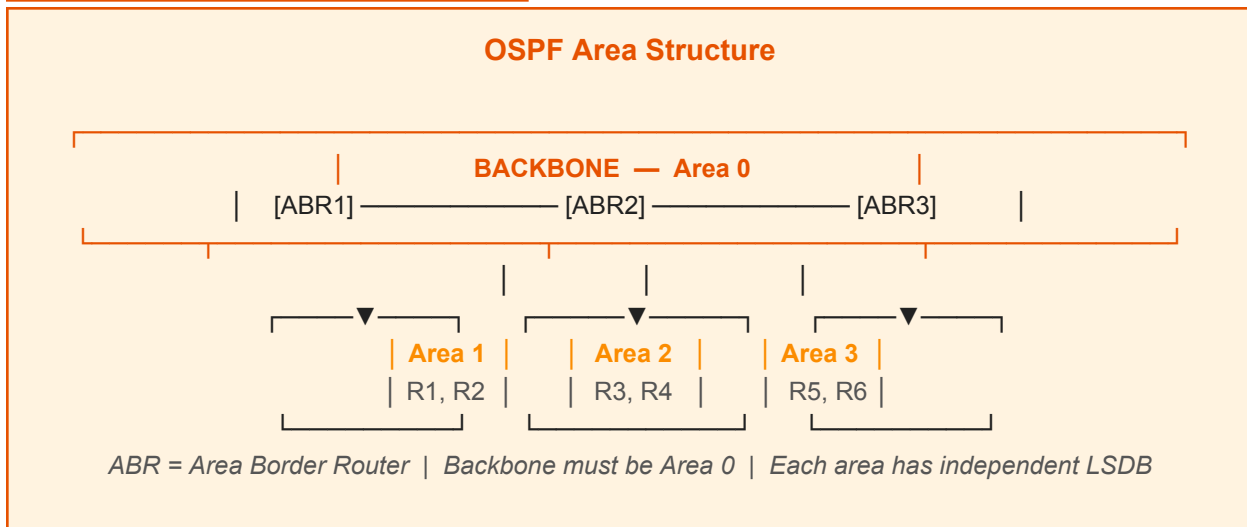
What is OSPF?

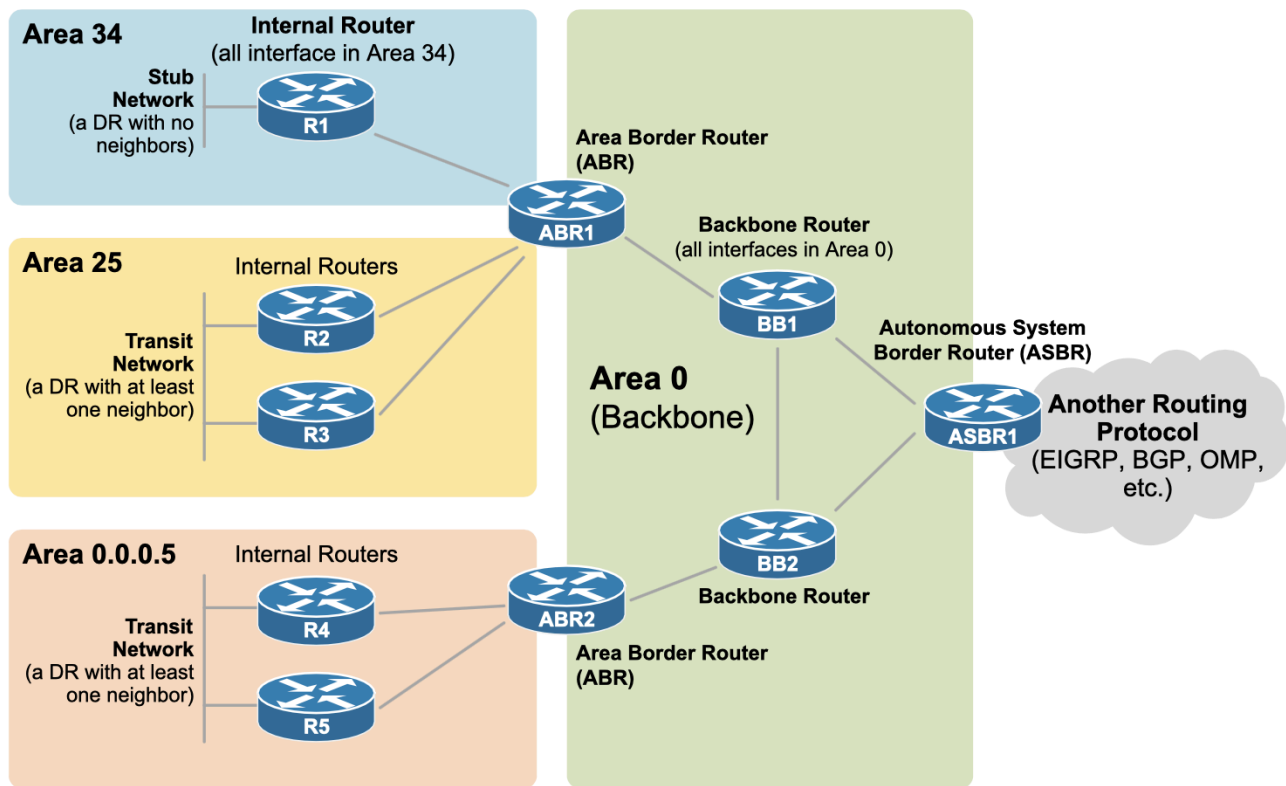
OSPF is a **link-state IGP** that runs **Dijkstra's Shortest Path First (SPF)** algorithm. Each router builds a **complete topology map (LSDB)** of the network and independently calculates the best path. It uses **cost (bandwidth-based)** as its metric. OSPF supports **Areas** for hierarchical scalability.

WORKING PHASES / NEIGHBOUR STATE MACHINE

State / Phase	Description & Example
DOWN	No Hello packets received yet. Initial state on startup.
INIT	Hello received from neighbour, but own Router ID not in the packet yet.
2-WAY	Bidirectional communication established. DR/BDR election happens here on multi-access networks. <i>Example: Routers R1 and R2 on the same Ethernet segment see each other's Hellos → 2-WAY.</i>
EXSTART	Master/Slave elected via highest Router-ID. DBD (Database Description) exchange begins.
EXCHANGE	Routers exchange DBD packets summarising their LSDB — like sharing a table of contents.
LOADING	Routers send LSR (Link State Request) for missing LSAs, receive LSU (Link State Update). <i>Example: R1 requests LSA for Network 10.1.1.0/24 it saw in R2's DBD but doesn't have.</i>
FULL	LSDB synchronised. SPF runs. Best routes installed in routing table. <i>Example: R1 calculates shortest path to 10.2.0.0/24 via R2 (cost 10+5=15) < via R3 (cost 30). Installs R2 as next-hop.</i>

OSPF HIERARCHICAL AREA DESIGN





✓ ADVANTAGES

- Fast convergence (SPF recalculates on topology change)
- Scalable via hierarchical areas
- Supports VLSM, CIDR, IPv4 & IPv6 (OSPFv3)
- **No hop-count limit**
- Equal-cost multi-path (ECMP) load balancing
- Authentication support (MD5, SHA)

✗ LIMITATIONS

- High memory/CPU usage for large LSDB
- Complex to configure and troubleshoot
- **All areas must connect to Area 0**
- SPF computation intensive on topology changes
- Requires careful IP addressing plan

4. BGP — Border Gateway Protocol (eBGP & iBGP)

What is BGP?

BGP is the **routing protocol of the Internet**. It is an **Exterior Gateway Protocol (EGP)** using a **path-vector algorithm**. BGP exchanges **Network Layer Reachability Information (NLRI)** over **TCP port 179**. It uses rich **path attributes (AS_PATH, NEXT_HOP, LOCAL_PREF, MED...)** for policy-based routing.

eBGP vs iBGP — KEY DIFFERENCES

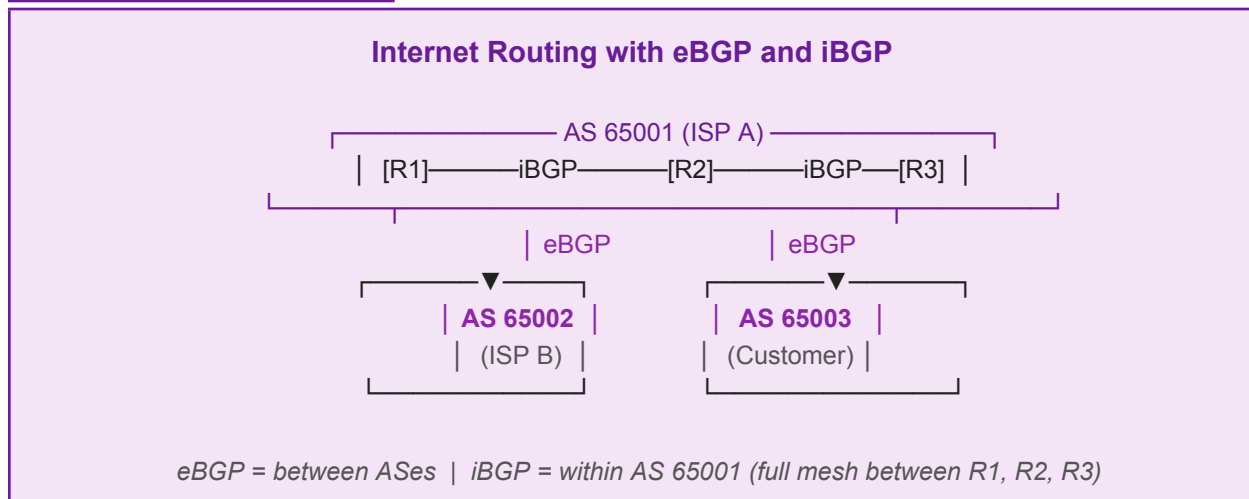
Property	eBGP (External BGP)	iBGP (Internal BGP)
Scope	Between different Autonomous Systems	Within the same AS

AS_PATH	Adds own AS number to path	AS_PATH NOT modified
TTL	Default TTL = 1 (must be adjacent)	Default TTL = 255 (any hop)
Next Hop	Next-hop changed to own address	Next-hop unchanged (IBGP peer must resolve)
Full Mesh	Not required	Required (or use Route Reflectors / Confederations)
Typical Use	ISP-to-ISP, peering at IXPs	Inside enterprise/ISP for consistent policy

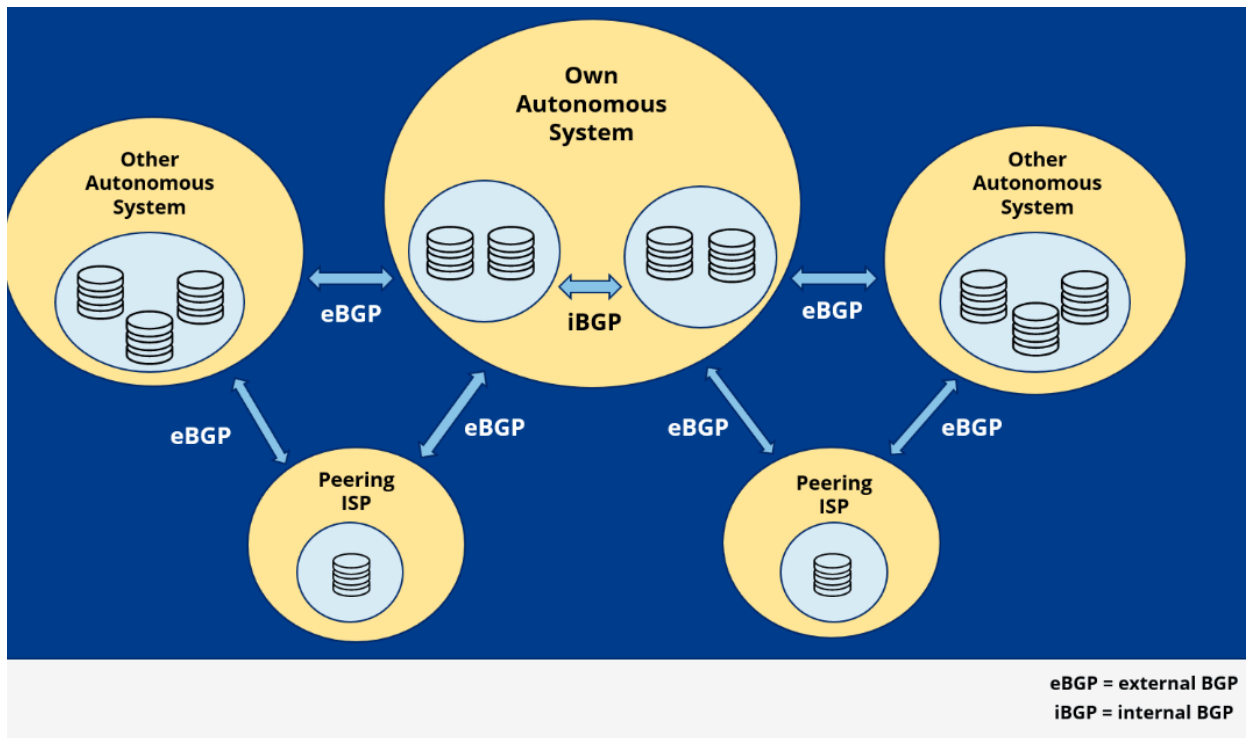
BGP SESSION ESTABLISHMENT PHASES

State	Description
IDLE	BGP waiting. TCP connection not yet initiated. Triggered by 'Start' event (e.g., 'neighbor' command).
CONNECT	TCP 3-way handshake in progress to peer on port 179.
ACTIVE	TCP handshake failed. BGP retries. If stuck here → check TCP reachability / ACLs.
OPEN SENT	TCP connected. BGP OPEN message sent (BGP version, AS number, Hold Time, Router-ID). <i>Example: AS 65001 sends OPEN to AS 65002.</i>
OPEN CONFIRM	OPEN received and validated. BGP waits for KEEPALIVE or NOTIFICATION.
ESTABLISHED	Session fully up. UPDATE messages exchanged (prefix + path attributes). <i>Example: AS 65001 advertises 203.0.113.0/24 with AS_PATH [65001] to AS 65002.</i>

BGP TOPOLOGY EXAMPLE



AS_PATH example: 203.0.113.0/24 advertised by AS 65003 → AS 65001 sees [65003] → AS 65002 sees [65001, 65003]



KEY BGP PATH ATTRIBUTES

AS_PATH	NEXT_HOP	LOCAL_PREF	MED
Loop prevention list of ASes	IP of next router for the prefix	Prefer exit within AS (higher = preferred)	Multi-Exit Discriminator: hints to neighbour AS on preferred entry

✓ ADVANTAGES

- Powers the entire Internet (700k+ routes)
- Rich policy control via path attributes
- Highly scalable — no hop-count limit
- Supports VLSM, CIDR, IPv4 & IPv6
- Stable, reliable TCP-based session
- Route filtering and traffic engineering

✗ LIMITATIONS

- Very complex to configure and tune
- Slow convergence (minutes) vs. IGP
- **iBGP requires full mesh (mitigated by RR)**
- High memory usage (full Internet table ~900k routes)
- **Misconfiguration can cause global routing outages**

PROTOCOL COMPARISON SUMMARY

Feature	AODV	RIP	OSPF	BGP
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Algorithm	Bellman-Ford	Bellman-Ford	Dijkstra SPF	Path Vector
Type	Reactive	Proactive	Proactive	Proactive
Metric	Hop count	Hop count	Cost (bandwidth)	Path attributes
Convergence	Fast (on-demand)	Slow (~mins)	Fast (<1 s)	Slow (~mins)
Scalability	Small MANETs	≤15 hops	Large enterprises	Internet-scale
Loop Prevention	Seq. numbers	Split Horizon	LSDB (SPF)	AS_PATH
Use Case	Wireless/mobile	Small LAN	Enterprise/ISP	Internet / EGP